

Technical Architecture — Phase 2 MVP Core

Commit: 0f0280d9d8b8ca95f539b09b41f6e960a1b0b42e **Deploy:** https://ratequip-web.vercel.app

Stack

- **App:** Next.js App Router (RateQuip web) on Vercel - **Auth:** Clerk (hosted sign-up / sign-in / verify / reset / MFA) - **Database:** Neon Serverless Postgres (rq schema, Drizzle migrations) - **Billing:** Stripe Test Mode (Checkout + webhooks → credit ledger) - **Runtime store:** Dual-path (runtime memory + optional Neon) for durability demos

Logical diagram

```
` Browser → Vercel (Next.js) | Clerk session | Server Actions / API routes (RBAC gated) | Neon Postgres (rq.*) | Stripe webhooks → billing ledger / wallet `
```

Tenant relationship (conceptual)

```
` User (Clerk) | Role: buyer | supplier | admin | Company membership / claim | Reviews (moderation queue) | RFQs / Quotes | Wallet + credit_ledger (tenant-scoped operations) `
```

Key code areas

Concern	Location
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Organic growth / company draft recovery	src/lib/organic-growth/store.ts, src/lib/actions/organic-growth.ts
RFQ validation / revise / award	src/lib/rfq/validation.ts, src/lib/actions/marketplace.ts
Reviews	src/lib/db/phase2.ts
Billing	src/lib/billing/operations.ts, /api/v1/billing/*, Stripe webhook
Health	GET /api/v1/health

Production health (verified)

demoMode: false, database: true` on https://ratequip-web.vercel.app/api/v1/health at package build time.